

MUHAMMAD ASAD SHAFIQ

Software Engineer

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About

Software Engineering student specializing in network security and cybersecurity, with practical experience building secure full-stack applications using encryption, authentication, and secure API design. Strong foundation in networking fundamentals and a continuous learner committed to a long-term career in cybersecurity.

Education

COMSATS University Islamabad, Lahore Campus

Bachelor of Science in Software Engineering

2024 – 2028

Lahore, Pakistan

Punjab Group of Colleges

Faculty of Science (FSc.)

The Educators

Matriculation

Areas of Interest

- Software Engineering
- Computer Networking
- Network Security
- Cloud Computing
- Cloud Security

Projects

LockBox — Zero-Knowledge E2EE Note & File Sharing Platform | *Node.js, Express.js, JavaScript, Web Crypto API*

- Developed a zero-knowledge note and file sharing platform with client-side AES-256-GCM encryption using the Web Crypto API, ensuring servers store only encrypted ciphertext.
- Implemented secure URL fragment key sharing (`#key`), PBKDF2-based password encryption (200,000 iterations), and burn-on-read functionality with automatic deletion and configurable auto-wipe timers.
- Applied security best practices including Redis TTL-based auto-expiry, filename sanitization, cryptographically secure passphrase generation (`crypto.getRandomValues()`), and AES-GCM authentication for tamper detection.

RideRevenue Tracker – Fleet Management System | *Node.js, Express.js, MySQL, JavaScript, RESTful API*

- Developed a full-stack fleet management platform for ride-sharing vehicle owners, featuring driver management, vehicle assignment, shift tracking, financial records, JWT authentication, and a normalized MySQL database with RESTful APIs.
- Implemented automated daily/monthly/custom-range earnings reports with fixed-salary and percentage-based compensation models, while enforcing security through bcrypt password hashing, input validation, SQL injection prevention, and role-based access control.

Technical Skills

Programming Languages: C++, Java, Python, Web Technologies

Networking: OSI Model, TCP/IP Stack, Subnetting, DNS, HTTP/S, Network Protocols

Core CS Concepts: Object-Oriented Programming (OOP), Data Structures & Algorithms, File Handling

Tools & Platforms: Git, GitHub, Linux, Cisco Packet Tracer, Wireshark

References

Available upon request.